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Hybrid renewable power supply for rural health clinics (RHC) in six geo-political zones of Nigeria

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ABSTRACT

The potentials of major renewable energy sources (wind and solar) in selected locations across the six geo-political regions of Nigeria, based on long-term daily meteorological data spanning between 18 and 39 years were reviewed in this study. In addition, the techno-economic feasibility of utilizing hybrid photovoltaic/wind/diesel with battery storage systems to meet the load of a typical rural healthcare facility at the selected sites were assessed. The optimum dimensions of the system are defined for the locations. Hybrid Optimization Model for Electric Renewable (HOMER) software developed by National Renewable Energy Laboratory's (NREL) was employed to conduct the study. Findings from the study showed that Sokoto and Jos exist in the high wind potential regions, while the remain sites are only suitable for small wind applications. Values obtained for global, beam and diffuse radiation as well as clearness index, show that all the sites enjoy considerable solar energy potential suitable for varying degree of solar energy applications. Monthly optimum tilt angle for Iseyin, Sokoto, Maiduguri, Jos, Enugu and Port-Harcourt lies in the range of 0–39.8°, 0–44.5°, 0–44.1°, 0–43.2°, 0–38.5° and 0–36.3° respectively thereby having the optimum angle to be equal to 0° in all sites in April, May, June, July and August. The simulation results from HOMER indicate that the hybrid system is the best option for all the sites considered in this study. The PV/wind/diesel/battery hybrid system configuration is considered optimum for RHC applications at Sokoto, Maiduguri, Jos and Enugu, while hybrid systems involving PV/diesel/battery is considered ideal for RHC at remote locations within Iseyin and Port-Harcourt, due to the quality of renewable energy potential. The diesel-only system provides the highest COE (\$0.911/kWh), and emits 9211 kg of CO₂ per year in all the site considered.

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Introduction

The reliability of electric power supply is considered a vital commodity towards the development of healthcare in rural/remote areas; the lack of which has limited its efficient delivery in many rural communities of the world, thus putting human lives at risks [1]. Modern advances in the distribution of vaccines and other cold-chain dependent drugs, as well as global push to deliver antiretroviral drugs (ARV) and services to HIV patients worldwide, have presented new demands for electricity in health centers where there are limited or even non-access to reliable power supply. Operators of rural healthcare facilities in Nigeria and

developing nations in general, are challenged with many problems, which have hindered effective delivery to the rural populace. For example, unreliable power supply can render cold-chain activities inoperable, while a healthcare facility without means of illumination (lighting) can keep patients that arrive late in the night for medical attention, to wait until the following morning before medical attention can be rendered.

Electricity crisis in Nigeria

Nigeria is made up of six geo-political zones with total land mass of 923,769 sq. km (98.5% highland and 1.5% lowland) [2]. According to United Nation, the percentage of rural population in Nigeria is about 50.4% with mere 36% of this population have access to electricity, and the majority with less than 4 h a day [3,4]. Commercial electricity generation in Nigeria currently comes from seven power stations and various independent power

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Nomenclature

RET	renewable energy technologies	$\Gamma(x)$	gamma function
RHC	rural health clinic	ρ	air density (kg/m ³)
HPS	hybrid power system	α	surface roughness coefficient
SEG	solar energy gain	$P(v)$	wind power (W)
NIMET	Nigerian metrological agency	$\bar{K}_T$	monthly clearness index
SS	south–south	$\bar{H}$	monthly global solar radiation (kWh/m ² /day)
SW	south–west	$\bar{H}_o$	monthly mean extraterrestrial radiation (kWh/m ² /day)
SE	south–east	$\bar{H}_b$	monthly mean beam radiation (kWh/m ² /day)
NE	north–east	$\bar{H}_d$	monthly mean diffuse radiation (kWh/m ² /day)
NW	north–west	$\bar{H}_T$	monthly mean global solar radiation on tilted surface (kWh/m ² /day)
NC	north–central	β_{opt}	optimum angle of inclination (°)
$f(v)$	Weibull probability density function		
k	Weibull shape parameter		
c	Weibull scale parameter (m/s)		

projects that spread around the country [5]. Hence, the current available generation capacity in the country stands at 3.9GW and having per capita capacity of 28.6 W [6]. This is totally inadequate even for domestic consumption in a country with an approximate population of 170 million and the escalating socio-economic activities. For the country to meet its energy needs, it will require per capita capacity of 1000 W or power generating capacity of 140GW as against the present capacity [7]. Consequently, availability of power in the country varies from about 27% to 60% of installed capacity, while the transmission and distribution losses accounted for about 28% of electricity generated [8,9]. Meanwhile, electricity is required for basic developmental services as well as economic growth. An area with no electricity access usually lack essential infrastructures (e.g. school, medical center, communication, portable water supply, etc.). The Human Development Index values in electrified communities, is found to be higher than non-electrified communities [10], and the absence of reliable energy supply has left the rural populace socially backward leaving their economic potentials untapped.

Since 1999 that marked the beginning of democratic dispensation in Nigeria, a number of reforms have been established to open up power sector to privatization thus improving energy supply infrastructure in order to cope with the fast growing population. Some of these frameworks are the National Energy Policy (NEP), the passage of the Electric Power Sector Reform Act (EPSRA), the establishment of the Nigerian Electricity Regulatory Commission (NERC) and the Rural Electrification Agency (REA); through these, complete government control of power generation and distribution was removed, while the activities of all stakeholders and issuance of operating licenses to various parties in the power sector were better managed [11]. These reforms led to the development of various independent power projects (IPPs) [12], to complement the existing power infrastructure across the country. Nevertheless, with these reforms, the provision of electricity to the rural communities has not improved due to grid unavailability to remote locations, and the general grid unreliability; other constraints may include inaccessible terrain, communities hostilities, and the high cost grid extension to the communities [12,13]. Considering the fact that up to 70% of the rural areas are still without any form of electricity [14], it is thus obvious that rural communities may not adequately benefit from power reforms unlike urban settlements. Since improving rural access to electricity through grid extension does not look promising at present due to the associated costs, it is thus imperative that a system of autonomous, off-grid power generation be established for rural communities in Nigeria. A solution that is based on renewable energy (RE) technologies,

due to the vast deposits of the resources, coupled with the associated environmental friendliness, would be a viable option more so that rural locations have lower electricity demand.

Renewable energy potential in Nigeria

Nigeria has vast deposit of renewable energy resources, ranging from hydropower, wind, solar and biomass [15], etc. An overview of literatures on rural electrification proves that a combination of these energy sources is one of the most effective solutions to meet electricity need of the rural communities. RE have been proved to provide quality and reliable electricity for different applications in the rural areas [16,17]. The two commonly explored RE resources are wind and solar.

A comprehensive study of available long-term solar insolation data and wind regime in a particular location is essential in designing and predicting energy output of the respective energy conversion devices; the in-depth knowledge will help in determining their suitability for any particular applications. Over the years, many researches have been carried out in different part of the world to assess the potentials of these resources for various applications. In the work of Besarati et al. [18], the potentials of solar radiations in different parts of Iran were assessed, and results showed greater potential for energy application in the central and southern part of the country. In a related study, El Ouderni et al. [19], assessed global solar radiation on hourly, daily, monthly and seasonal basis to determine the suitability for power application at Borj-Cedrin in the gulf of Tunis in Tunisia; the performance of the developed experimental model was validated with the actual measured data, and results showed good agreement. A study on optimal tilt angle for solar collector orientation to enhance energy production was carried out at a location in Saudi Arabia in the work of Benghanem [20]. The results of the study showed a gain of 8% at a tilt of solar collector in a particular angle on monthly basis as compared to yearly tilt.

Studies on wind speed assessment from various locations around the world have also shown encouragements for power applications. In a recent work, Islam et al. [21], assessed wind power characteristics and potentials for power generation at two sites in Malaysia with wind data measured at 10 m hub height at the respective sites; maximum monthly wind speeds of 4.8 and 4.3 m/s, and power density of 67.4 and 50.8 W/m² were observed at Kudat and Labuan respectively. Also, in a related study by Mohammadi and Mostafaeipour [22], an assessment of the possibility of wind energy for power application at a particular location in Kurdistan provinces on hourly, daily, monthly, seasonal and

annual basis were conducted using Weibull distribution function. Outcome of the study show the site to be marginally suitable for wind power application.

In Nigeria, several research works have also been carried out on solar and wind energy assessment for power generation, but with no particular considerations for rural healthcare application. In Ohunakin et al. [23], solar applications and developments in Nigeria, were comprehensively discussed; various irradiation levels as distributed among the 36 States of the federation were divided into zones. Furthermore, since there are limited meteorological sites in the country, several empirical models had been developed for many locations across Nigeria, to predict global and diffuse radiation data, needed for various solar applications [24–33]. In wind availability assessment, Nigeria was found to be a poor/moderate wind regime with wind speeds ranging from 2.12 to 4.13 m/s in the southern part of the country with the exception of the coastal regions/offshore, and the north where wind speeds were found to range from 4.0 to 8.60 m/s [34–37].

The aim of this study is to analyze long-term meteorological data of six selected locations across the six geo-political regions of Nigeria to assess the techno-economic feasibility of utilizing hybrid PV–wind–diesel–battery power systems to meet the load of a typical rural health care (RHC) facility at remote locations, with lowest cost of energy. National Renewable Energy Laboratory's (NREL) Hybrid Optimization Model for Electric Renewable (HOMER) software has been employed to carry out the present study. The optimum dimensions of the system are defined for the six sites. Conclusions on the suitability of each of these renewable sources for rural healthcare applications in each of selected locations were drawn based on the results of analyses conducted.

Methodology

The sites were selected from each of the six geo-political zones of the country with a site representing each of the zones (Fig. 1).

The locations are characterized by different climatic conditions and include: Iseyin (south–west), Sokoto (north–west), Maiduguri (north–east), Jos (north–central), Enugu (south–east) and Port-Harcourt (south–south). The potential of solar energy resources at the selected sites were analyzed based on certain key solar resources parameters, such as monthly and annual global solar radiation (GSR), beam radiation, diffuse radiation, as well as clearness index. The optimal tilt angle for south facing solar collector orientation was also determined. Wind energy potentials on the other hand were examined based on monthly mean daily wind speed data.

Study sites and data description

The meteorological data (wind speed and solar radiation) used in the analysis were obtained from the Nigerian Metrological Agency (NIMET), located in Oshodi, Lagos [38]. Archived daily wind speed data spanning over a period between 28 and 39 years were captured at 10 m height by a cup-generator anemometer at all the locations, while the daily global solar radiation on horizontal plane were measured using Gunn-Bellani radiometer, with readings ranging between 18 and 31 years. These data were computed as average of the data for each month. The geographical information of the selected locations as well as period of data collection are presented in Table 1.

Wind energy assessment

Over the years, various statistical models such as normal, log-normal, Weibull and Rayleigh distributions have been widely used for wind data analysis [39–43]. Among these statistical distributions, two-parameter Weibull probability distribution function was widely adopted for wind data assessments because it gives a good match with experimental data when compared to the other statistical models [35]. It is a special case of generalized gamma

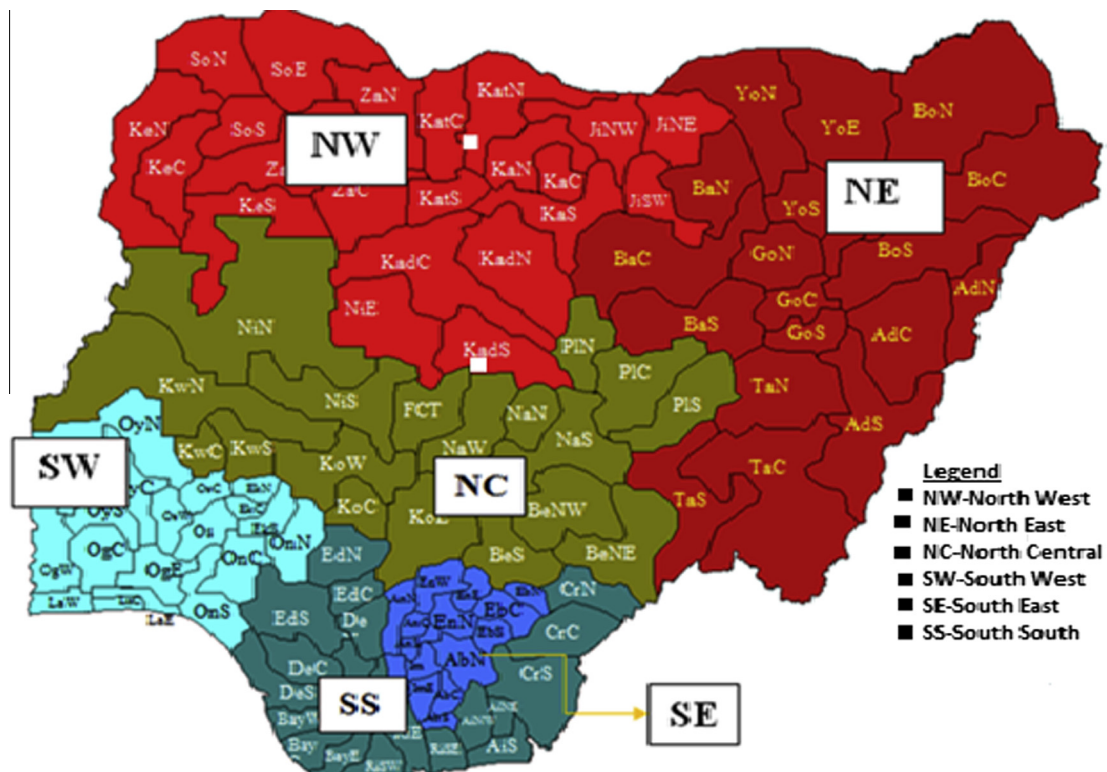


Fig. 1. Map of Nigeria representing the six geo-political zones.

Table 1

Characterization of the selected locations.

S/N	Location	Zone	Latitude (°N)	Longitude (°E)	Altitude (m)	Data period (wind speed)	Data period (solar radiation))
1	Iseyin	SW	7.96	3.60	330	(1983–2010)	(1990–2010)
2	Sokoto	NW	12.28	4.13	220	(1971–2009)	(1988–2008)
3	Maiduguri	NE	11.83	13.15	353.8	(1976–2007)	(1977–2007)
4	Jos	NC	9.92	8.9	1217	(1971–2006)	(1987–2007)
5	Enugu	SE	6.45	7.5	247	(1971–2007)	(1990–2007)
6	Port-Harcourt	SS	4.78	7.0	465	(1971–2005)	(1990–2007)

function commonly used in describing wind speed frequency distribution as well as the estimation of wind power density [44]. In this study, two-parameter Weibull distribution function (WDF) is employed to describe the monthly wind speed variation, wind seasonal changes and wind power density. The probability density function (PDF) for Weibull, can be expressed as [44]:

$$f(v) = \frac{k}{c} \left(\frac{v}{c}\right)^{k-1} \exp \left[-\left(\frac{v}{c}\right)^k \right] \quad (1)$$

while the corresponding cumulative probability function (CPF) is given in [35] as:

$$F(v) = 1 - \exp \left[-\left(\frac{v}{c}\right)^k \right] \quad (2)$$

where v is the wind speed, while k and c are known as shape and scale parameters respectively. These parameters can be calculated using different approaches such as the graphical method [45], standard deviation method [46], maximum likelihood method [47], power density method [48], etc. In this study, standard deviation method as expressed in Eqs. (3) and (4) was adopted [46,49]:

$$k = -\left(\frac{\sigma}{\bar{v}}\right)^{-1.086} \quad \text{for } 1 \leq k \leq 10 \quad (3)$$

$$c = \frac{\bar{v}}{\Gamma(1 + 1/k)} \quad (4)$$

$\bar{v}$ is the average wind speed in (m/s), σ is the standard deviation which shows degree of variation of wind speed and $\Gamma(x)$ is the gamma function which is given by:

$$\Gamma(x) = \int_0^\infty \exp(-u) u^{x-1} du \quad (5)$$

The estimation of wind power in terms of Weibull parameter is given by the following equation [50]:

$$P_d = \frac{P(v)}{A} = \frac{1}{2} \rho \int_0^\infty v^3 f(v) dv = \frac{1}{2} \rho c^3 \Gamma\left(1 + \frac{3}{k}\right) \quad (6)$$

where $P(v)$ is the wind power in (W), P_d is the wind power density (Wh/m²), A is the swept area of the wind turbine rotor (m²) and ρ is the air density of the selected sites (assumed to be 1.225 kg/m³ for all the sites considered in this study).

The corresponding mean energy density of the wind is obtained by multiplying Eq. (6) with period of time (T) under consideration [51]:

$$E_d = \frac{1}{2} \rho c^3 \Gamma\left(1 + \frac{3}{k}\right) T \quad (7)$$

The annual energy output is given in [35] as:

$$\bar{E}_a = \sum_{n=1}^{12} \bar{E}_{jm} \quad (\text{kWh/m}^2/\text{year})$$

where, $\bar{E}_{jm}$ is the extractable mean monthly energy output given by $24 \times 10^{-3} P_d \times T$, with P_d given by Eq. (6) and T is the number of day

in the month considered. Furthermore, exploration of wind speed at different hub height as a function of reference height can be obtained using the relation [52]:

$$\frac{v}{v_0} = \left(\frac{h}{h_0}\right)^\alpha \quad (8)$$

where v is the wind speed at required hub height ' h ', v_0 is the wind speed at reference height h_0 and α is site surface roughness coefficient with value ranges from 0.10 for flat surfaces and more than 0.25 for heavily forested landscape. The typical value of 0.14 for low roughness surfaces as discussed in Masters [53], is use in this study.

Solar energy assessment

Characteristics of solar energy resources at selected sites are analyzed based on average monthly global solar radiation as well as monthly clearness index, monthly average daily diffuse and beam radiations. They are very essential for the purpose of efficient design and performance evaluation of solar energy applications. For all the sites considered in this study, the available daily data were averaged to obtain the monthly and annual mean values.

For the monthly analysis employed in this study, monthly mean clearness index ($\bar{K}_T$) defined in terms of monthly mean global solar radiation incident on the horizontal surface ($\bar{H}$), and the corresponding monthly mean extraterrestrial radiation ($\bar{H}_o$) is expressed as [54]:

$$\bar{K}_T = \frac{\bar{H}}{\bar{H}_o} \quad (9)$$

where $\bar{H}_o$ is expressed as [25]:

$$\bar{H}_o = \frac{24}{\pi} G_{sc} \left(1 + 0.033 \cos \frac{360d}{365} \right) \times \left(\frac{\pi w_s}{180} \sin \varphi \sin \delta + \cos \varphi \cos \delta \sin w_s \right) \quad (10)$$

G_{sc} is the solar constant given as 1367 W/m², φ is the latitude of the site considered, w_s is the monthly average sunrise hour angle, d is average day of the month, and δ is the declination angle expressed by Eq. (11) as:

$$\delta = 23.45 \sin \left(\frac{360(284 + d)}{365} \right) \quad (11)$$

The monthly mean sunrise hour angle (w_s) is obtained using Eq. (12):

$$w_s = \cos^{-1}(-\tan \varphi \tan \delta) \quad (12)$$

Total solar radiation on horizontal surface

The total monthly solar radiation on horizontal surface is a composite of monthly mean beam ($\bar{H}_b$) and diffuse radiation incident on horizontal surfaces ($\bar{H}_d$) and represented in Eq. (13):

$$\bar{H} = \bar{H}_b + \bar{H}_d \quad (13)$$

Meanwhile, monthly mean diffuse radiation on horizontal surfaces ($\bar{H}_d$) can be estimated by a correlation proposed by Erb et al., [55] as:

$$\bar{H}_d = \begin{cases} 1.391 - 3.560\bar{K}_T + 4.189\bar{K}_T^2 - 2.137\bar{K}_T^3, & \text{for } w_s \leq 81.4^\circ \text{ and } 0.3 \leq \bar{K}_T \leq 0.8 \\ 1.311 - 3.022\bar{K}_T + 3.427\bar{K}_T^2 - 1.821\bar{K}_T^3, & \text{for } w_s \geq 81.4^\circ \text{ and } 0.3 \leq \bar{K}_T \leq 0.8 \end{cases} \quad (14)$$

Total radiation on tilted surface

For the purpose of solar energy design and performance calculation, it is often necessary to calculate solar radiation on the tilted surfaces, from the measured global solar radiation data. Solar radiation on tilted surface is a composite of beam and diffuse radiation on tilted surface; including ground reflected radiation. Several models have been proposed by researchers to estimate global solar radiation on tilted surface [56–58]. The most prominent is the isotropic model proposed by Hottel and Woertz [59], and later modified by Liu and Jordan [60] as expressed in Eq. (15) as:

$$\bar{H}_T = \bar{H}_b \bar{R}_b + \bar{H}_d \bar{R}_d + \bar{H}_g \rho_g \left(\frac{1 - \cos \beta}{2} \right) \quad (15)$$

where β is the tilt angle of the solar collector (whose optimum value is to be determined) as shown in Fig. 2, while ρ_g represent the ground albedo and taken as 0.2 in this study [54]. $\bar{R}_b$ is the ratio of the monthly beam radiation on inclined surface to the horizontal surface as given in [54]:

$$\bar{R}_b = \frac{\cos(\varphi - \beta) \cos(\delta) \sin(w'_s) + (\pi/180) w'_s \sin(\varphi - \beta) \sin(\delta)}{\cos(\varphi) \cos(\delta) \sin(w_s) + (\pi/180) w_s \sin(\varphi) \sin(\delta)} \quad (16)$$

w'_s represent the sunrise hour angle on tilted surface, given in Eq. (17) with “min” denoting the smallest of the items in the bracket.

$$w'_s = \min \left\{ \begin{array}{l} \cos^{-1}(-\tan \varphi \tan \delta) \\ \cos^{-1}(-\tan(\varphi - \beta) \tan \delta) \end{array} \right\} \quad (17)$$

$\bar{R}_d$ (Eq. (18)) is ratio of monthly mean diffuse solar radiation on the tilted surface to the measured global radiation on horizontal surface. The Hay-Davies model [58], which takes into account the circumsolar components of diffusion (anisotropy index), is adopted in this study to determine the value of $\bar{R}_d$.

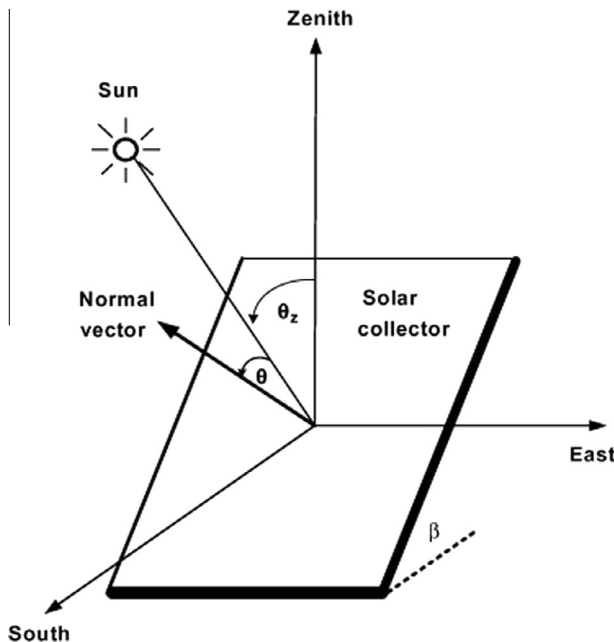


Fig. 2. Schematic view of solar collectors with the characteristic angles.

$$\bar{R}_d = \frac{\bar{H}_b}{\bar{H}_o} \bar{R}_b + \left(1 - \frac{\bar{H}_b}{\bar{H}_o} \right) \left(\frac{1 + \cos \beta}{2} \right) \quad (18)$$

Optimum tilt angle is a function of geographical latitude and local climatic condition of any site under consideration; it is the angle of inclination that produced maximum monthly mean global radiation according to Eq. (15). The optimum value is estimated separately for each of the studied sites. The tilt angles examined in this study varies between 0° and 90° and codes written in MATLAB were used to determine the optimum value for each respective months in all the selected sites.

The two prevailing seasons in Nigeria (dry and rainy seasons) were considered in the computations of the seasonal optimal angle. The rainy season months range between April and September in the northern part of the country, and between April and October in the south region. On the other hand, dry season months extends from October to March in the northern region, and November to March in the south. The amount of energy gain as a result of the optimal tilt orientation of solar collector, compared to collector positioned in horizontal plane is also considered in this analysis. The percentage of solar energy gain (SEG) as a result of optimal tilt for each months of the year, as well as for the complete year is obtained using Eqs. (19) and (20) respectively.

$$SEG = \frac{\bar{H}_T - \bar{H}}{\bar{H}} \times 100 \quad (19)$$

$$SEG = \frac{\sum_{i=1}^{12} \bar{H}_T - \sum_{i=1}^{12} \bar{H}}{\sum_{i=1}^{12} \bar{H}} \times 100 \quad (20)$$

Assessment of energy demand of rural health clinics

An assessment of energy demands of any health facility is critical in the selection of the most suitable energy resources. For instance, a health post, which is the most basic health facility available in remote villages, is usually characterized with medical services such as: (i) treatment of minor illnesses or injuries, (ii) child deliveries and provision of basic immunization services. Energy need of such facilities can be considered to be relatively low, due to limited availability of medical equipment. On the other hand, a health clinic that offers wider range of medical services than health post, and possess equipment that allows more medical diagnoses can be consider high in energy demands. According to the United States Agency for International Development (USAID) [61], healthcare facility can generally be categorized into three. The first (Category A), is applicable to rural setting and characterized with limited medical services and staff. In this category, electricity is often required for lighting during evening hours operations and to support limited surgical procedures (such as suturing), refrigeration for maintenance of cold-chain vaccines and other medical supplies. Other basic equipment in this facility may include a centrifuge, hematology mixer, microscope, incubator and hand-held power aspirator. The estimated load demand for this category range between 5 and 10 kWh/day. The second category (Category B), contains medical equipment that are similar to the first category, but can as well accommodate sophisticated medical diagnostic equipment with frequent usage when compared to facility in first category. Additional energy consuming appliances that can be obtained in this category includes: refrigerators often used for food/blood bank storage, communications equipment to share information with medical expert in referral centers. The estimated energy demand in this category is in the range of 10–20 kWh/day. The third (Category C) usually serves as a primary referral center, because it can coordinate communication between several smaller facilities and major hospital in large cities.

This may also contain sophisticated diagnostic equipment (such as X-ray machine, CD4 counters, etc.) requiring additional power. Daily typical energy demand of these facilities is in the range of 20–30 kWh. However, majority of health centers in the studied location can be classified as Category A, hence the proposed energy resources have the capability of meeting the load demands. Typical load description and estimated demand of such facility is presented in Table 2.

Energy demand model

Hybrid Optimization Model for Electric Renewables (HOMER), a renewable energy-based system optimization tool developed by the National Renewable Energy Laboratory (NREL), was employed in modeling the energy system for technical and economic feasibility [62,63]. It is a general-purpose system design tool for electric power systems. Inputs to the software includes; site load profile, renewable energy resources data, fuel price, system control parameters, constraint parameter as well as component's technical and economics details. HOMER performs numbers of hourly simulations that ensure best possible matching between the load and the supply, for effective design of an optimal system, and thereafter creates a list of feasible system sorted according to the lowest net present cost (NPC). Proper combination of different energy sources can improve overall system reliability and efficiency. It can also reduce the energy storage requirement as compared to system consisting of only single renewable energy source [64,65]. With the complementary nature of wind and solar energy resources, hybrid wind/solar configuration with energy storage system offer a highly reliable source of power, which is suitable for electrical loads that need higher reliability (e.g. health services). A diesel generator can be added as a backup energy sources in the event of insufficient RE resources.

In a one-source renewable system, the system tends to be oversized to accommodate the load demand with resultant high capital cost; this is not the case in hybrid system because the various connected energy sources, complement one another by utilizing their individual capacities and improving the individual generators load factor [66]. The sizing of different components need to be considered in order to meet the load demand. These components include: PV array, wind turbine, diesel generator, converter, and batteries.

Stand-alone diesel system, hybrid PV–diesel system with battery, hybrid wind–diesel system with battery, PV–wind–diesel system with battery configurations were analyzed in this study. The economic feasibility of each configuration is based on the net present cost (NPC); two other parameters including levelized cost of energy (COE) and renewable fraction (RF) were also employed as performance criteria for the selection of the optimum

configuration. CO₂ emissions (tons/year) and diesel consumption (l/year) were considered for the analysis of environmental friendly solution. All the input parameters are fed into the HOMER to determine the optimal system from the various configurations. The amount of power generated by each configuration, were compare to the annual load demands to ensure it meets the load demands. The model flowchart for the selection of different energy resources for RHC in each of the location is shown in Fig. 3.

Results and discussion

Wind speed frequency distribution

The annual probability density and cumulative distribution functions obtained for wind speed at the six selected locations are shown in Figs. 4 and 5 respectively. It can be observed from Fig. 4 that, the peak of the density functions of all the sites skewed towards the higher values of mean wind speed; the peak further indicates the most frequent velocity. The most frequent wind speeds of 9, 8 and 6 m/s are expected in Jos, Sokoto, and Maiduguri respectively (Fig. 4), while the least occurs in Port-Harcourt at 3.5 m/s. It can further be observed that Jos has the highest spread of wind speeds among the locations. Similar trend is observed in the cumulative probability distributions of wind speed in all the locations as shown in Fig. 5. For wind speeds greater or equal to 3.0 m/s cut-in wind speed, Sokoto, Jos, Enugu, Maiduguri, Port Harcourt, and Iseyin have frequencies of about 99.8%, 99.7%, 94.5%, 90.3%, 65.9% and 64.3% respectively.

All the sites have monthly peak frequency range between 26% and 40.6%, and having 26%, 28.5%, 35.3%, 37.3%, 40.2% and 40.6% for Iseyin, Jos, Maiduguri, Enugu, Sokoto and Port Harcourt respectively. Iseyin has its peak frequency at 4 m/s in December, Jos at 10 m/s in May, Maiduguri in September at 5 m/s, Enugu in October at 5 m/s, Sokoto at 7 m/s in August and Port Harcourt in January at 4 m/s.

Mean wind speed and mean power density

The monthly mean wind speeds at 10 m height, wind power density and Weibull parameters for the selected sites are presented in Tables 4 and 5. The monthly mean wind speed varies between 3.48 m/s in November and 4.71 m/s in April for Iseyin. The shape parameter k varies between 2.02 and 2.92, while the scale parameter c lies between 3.92 and 5.27 m/s. Power density varies between 46.48 W/m² in October and 89.69 W/m² (March). The annual mean wind speed and wind power density are found to be 4.158 m/s and 65.36 W/m². According to the wind power classification scheme proposed by Elliot and Schwartz [67] of Pacific Northwest Laboratory (Table 3), the monthly mean as well as

Table 2
Typical load description and estimated demand of rural health clinic.

Load description	A Quantity	B Power (W)	C = A × B Total (W)	D On-time (h/d)	E = C × D/1000 (kWh/day)
Lighting	10	10	100	2–10	0.2–1.0
Vaccine refrigerator	1	60	60	6–12	0.36–0.72
Small refrigerator	1	300	300	5	1.5
Centrifuge	1	575	575	1–1.5	0.575–0.8625
Hematology mixer	1	28	28	1–1.5	0.028–0.042
Microscope	2	15	30	3–4	0.045–0.12
Incubator	1	400	400	2–12	0.8–4.8
Water bath	1	1000	1000	1–2	1–2
Communication	1				
Stand-by		2	2	12	0.024
Transmitting		30	30	1–2	0.03–0.06
Total			2525		4.4–10.3

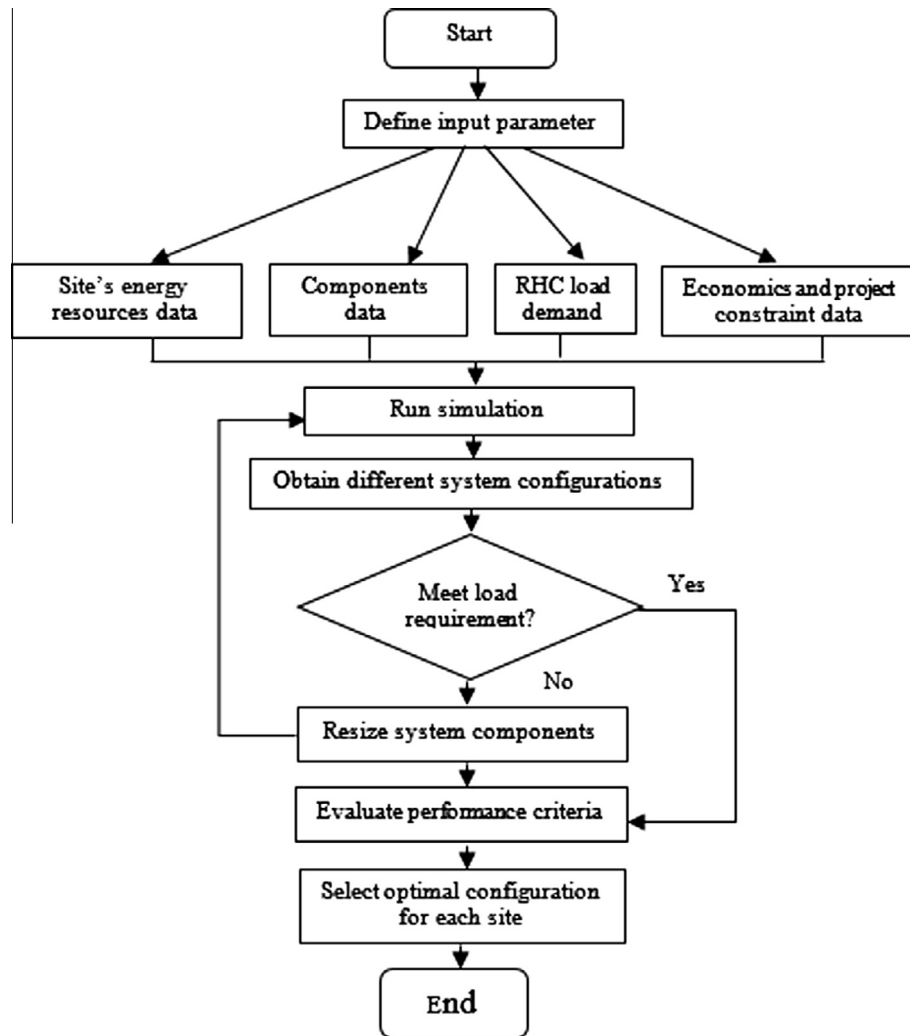


Fig. 3. Energy system configurations selection flowchart.

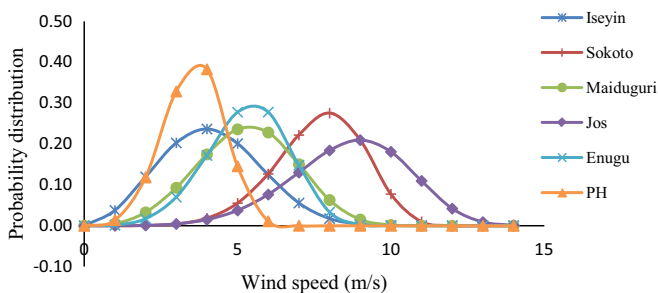


Fig. 4. Probability density function of the selected sites.

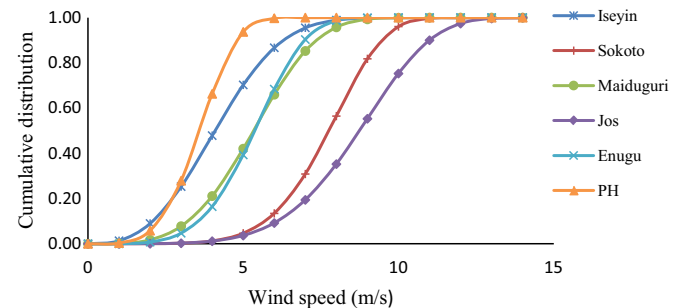


Fig. 5. Cumulative density function of the selected site.

the annual mean power density for Iseyin falls into class 1, and may only be applicable for battery charging, water pumping and other small wind power applications.

The minimum and maximum values of the monthly mean wind speeds are found to be 5.81 and 9.03 m/s respectively for Sokoto. The monthly mean shape parameter (k) has values between 3.171 and 9.938, while the scale parameter c is between 6.303 and 9.719 m/s. The monthly mean wind power density varies from 134.79 W/m² in September to 494.61 W/m² in January. However, the annual mean power density for this site stand at 321.53 W/m², which makes the site to ranked at Class 6, and thus

Table 3
Wind power classification [67].

Power Class	Power density (W/m ²) at 10 m
1	$0 < P \leq 100$
2	$100 < P \leq 150$
3	$150 < P \leq 200$
4	$200 < P \leq 250$
5	$250 < P \leq 300$
6	$300 < P \leq 400$
7	$400 < P \leq 1000$

Table 4

Monthly wind variation of Weibull parameters for the selected sites.

Site	Iseyin		Sokoto		Maiduguri		Jos		Enugu		Port Harcourt	
	<i>k</i>	<i>c</i> (m/s)	<i>k</i>	<i>c</i> (m/s)	<i>k</i>	<i>c</i> (m/s)	<i>k</i>	<i>c</i> (m/s)	<i>k</i>	<i>c</i> (m/s)	<i>k</i>	<i>c</i> (m/s)
Jan	3.182	4.702	5.262	9.719	3.185	5.517	3.496	10.17	3.485	6.248	4.185	3.741
Feb	3.257	4.864	4.264	9.486	3.587	6.549	4.73	10.008	4.86	6.175	4.623	4.256
Mar	2.504	5.117	5.459	7.965	3.764	6.753	6.602	9.694	5.742	6.807	5.439	4.227
Apr	3.005	5.274	5.734	8.202	4.251	6.377	3.944	9.882	5.09	6.767	4.907	4.339
May	3.095	4.842	9.938	9.147	3.976	6.412	7.553	9.595	4.426	5.868	4.656	4.057
Jun	2.722	4.89	9.936	9.494	3.585	6.727	5.405	9.281	4.838	5.685	4.149	4.041
Jul	2.891	4.711	6.534	8.519	3.029	6.336	7.116	9.239	5.45	5.939	3.461	4.081
Aug	2.916	4.776	7.537	6.635	3.866	5.107	6.217	8.854	3.937	6.007	4.281	4.319
Sep	2.445	4.443	5.351	6.303	4.796	4.902	4.611	7.824	4.819	5.294	4.617	3.983
Oct	2.387	4.073	3.171	6.724	3.742	5.039	4.406	8.382	5.253	4.952	3.487	3.546
Nov	2.022	3.927	3.964	8.279	3.546	5.664	4.023	9.419	3.75	4.562	3.452	3.237
Dec	3.027	4.377	5.872	8.483	3.441	5.206	4.725	10.228	3.341	5.515	2.879	3.208

Table 5

Mean monthly and annual variation of wind speed characteristics of the selected sites at 10 m height.

Site	Iseyin		Sokoto		Maiduguri		Jos		Enugu		Port Harcourt	
	<i>V</i> (m/s)	<i>P</i> (W/m ²)	<i>V</i> (m/s)	<i>P</i> (W/m ²)	<i>V</i> (m/s)	<i>P</i> (W/m ²)	<i>V</i> (m/s)	<i>P</i> (W/m ²)	<i>V</i> (m/s)	<i>P</i> (W/m ²)	<i>V</i> (m/s)	<i>P</i> (W/m ²)
Jan	4.21	61.44	8.95	494.61	4.940	99.22	9.150	603.38	5.620	140.00	3.400	28.90
Feb	4.36	67.47	8.63	469.62	5.900	159.99	9.160	544.57	5.660	127.55	3.890	41.98
Mar	4.54	89.21	7.35	271.71	6.100	173.38	9.040	488.07	6.300	169.32	3.900	40.62
Apr	4.71	88.69	7.59	296.17	5.800	142.71	8.950	538.06	6.220	167.32	3.980	44.24
May	4.33	67.82	8.70	415.45	5.810	146.76	9.010	474.19	5.350	110.60	3.710	36.34
Jun	4.35	74.12	9.03	464.55	6.060	173.39	8.560	430.12	5.210	99.60	3.670	36.44
Jul	4.20	64.29	7.94	331.25	5.660	153.24	8.650	422.90	5.480	112.65	3.670	39.11
Aug	4.26	66.75	6.23	156.81	4.620	74.55	8.230	371.96	5.440	120.89	3.930	44.29
Sep	3.94	59.35	5.81	134.79	4.490	63.90	7.150	260.83	4.850	80.43	3.640	34.40
Oct	3.61	46.48	6.02	179.97	4.550	72.12	7.640	322.51	4.560	65.44	3.190	25.60
Nov	3.48	48.18	7.50	316.03	5.100	103.84	8.540	464.10	4.120	53.50	2.910	19.52
Dec	3.91	50.54	7.86	327.40	4.680	81.31	9.360	581.19	4.950	97.52	2.860	20.35
Annual	4.16	65.36	7.63	321.53	5.309	120.37	8.620	458.49	5.310	112.07	3.563	34.32

applicable for grid connected wind power applications. In the case of Maiduguri, the minimum and maximum values of the monthly mean wind speeds are 4.49 m/s (in September) and 6.10 m/s (in March) respectively. The monthly mean power density varies from 63.9 W/m² in September to 173.38 W/m² in March; with the annual mean power density for the site as 120.37 W/m², the location is found to rank in Class 2 (Table 3). The minimum and maximum values of the monthly mean wind speeds varies from 7.15 (September) to 9.36 m/s (December) respectively in Jos. The monthly mean shape parameter *k* ranges between 3.496 and 7.553, while the scale parameter *c* is from 7.824 and 10.228 m/s. Monthly mean wind power density varies from 260.83 W/m² in September to 603.38 W/m² in January. The average annual power density stands at 458.89 W/m², making the site to fall in Class 7 of the wind power classification. The classification places Jos on a high potential site for large-scale grid connected wind turbine applications. The wind speed characteristic of Enugu and Port-Harcourt are also shown (Tables 4 and 5).

The minimum and maximum values of wind speed are 4.120 and 6.300 m/s respectively for Enugu whereas they exist as 2.860 and 3.980 m/s in Port-Harcourt for the minimum and maximum. Considering the wind power density, the dry season is found to be the windiest season for both sites. Monthly wind power ranges from 53.5 W/m² in November to 169.32 W/m² in March at Enugu, while it varies from 19.52 to 44.29 W/m² in November and August respectively in Port-Harcourt. With the wind power classification in Table 3, Enugu falls under Class 2, and Port-Harcourt in Class 1, thereby making the locations to be marginally considered for wind power applications.

Monthly and yearly clearness index

The monthly mean clearness index for all the sites are shown in Fig. 6. It varies from 0.28 to 0.56 for Iseyin, 0.42 to 0.61 for Sokoto, 0.46 to 0.63 for Maiduguri, 0.41 to 0.65 for Jos, 0.34 to 0.56 for Enugu, and 0.36 to 0.52 for Port-Harcourt. Since all the sites considered in this study experience different weather conditions, the amount of solar radiations that pass through the atmosphere as well as the characteristic of solar radiation, changes from one sites to another. As a result, the monthly clearness index varies from one zone to another. However, based on the obtained clearness index, it can be observed that all the sites considered enjoyed clear sky conditions throughout the year except for Port-Harcourt, Enugu and Iseyin which had some months (July, August and September) with rain cloudy sky.

Solar radiation on horizontal, tilted surfaces and optimum tilts

The monthly mean beam ($\bar{H}_b$) and diffuse ($\bar{H}_d$) for each site were computed from measured global solar radiation ($\bar{H}$) incident on the horizontal surface based on Eqs. (13) and (14). These parameters are required in assessment study of any solar energy applications. Fig. 7 depicts the mean monthly radiation for each sites. Maiduguri is seen to have the highest global solar radiation among the selected sites, while the least occurred at Port Harcourt. The maximum and minimum values for Maiduguri were obtained in April and August respectively, while the maximum and minimum value was obtained in December and July respectively in Port Harcourt.

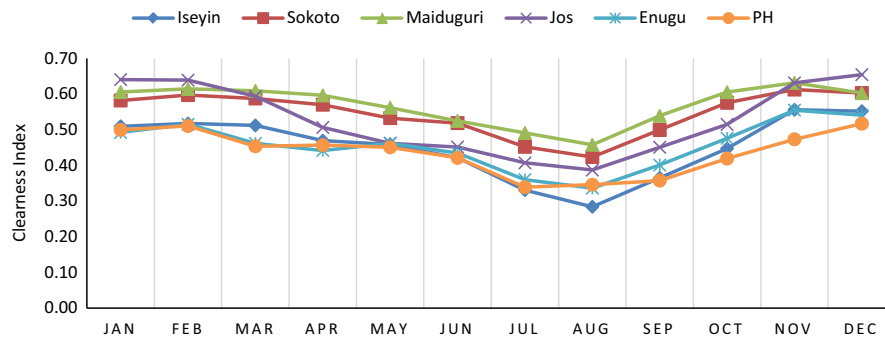


Fig. 6. Monthly variation in clearness index for the six locations.

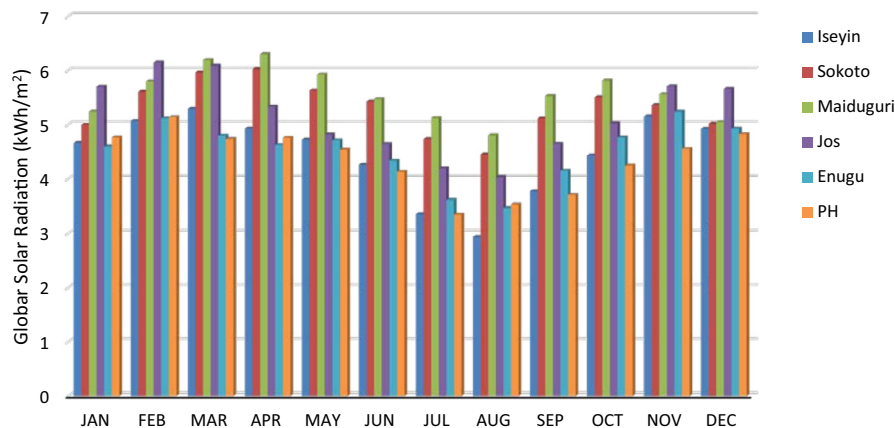


Fig. 7. Mean monthly solar radiation on horizontal plane.

Table 6

Annual beam, diffuse, horizontal and tilted radiation for the selected sites.

Station/parameters	$\bar{H}_0$ (kWh/m ²)	$\bar{H}_d$ (kWh/m ²)	$\bar{H}$ (kWh/m ²)	$\bar{H}_T$ (kWh/m ²)
Iseyin	2.364	2.090	4.459	4.758
Sokoto	3.274	2.049	5.319	5.797
Maiduguri	3.544	2.023	5.568	6.060
Jos	3.122	2.045	5.167	5.640
Enugu	2.407	2.121	4.528	5.795
Port Harcourt	2.228	2.128	4.356	4.554

Values obtained for the remaining sites can also be seen in the figure. It can further be observed from results obtained, that all locations have higher levels of horizontal global solar radiations during the dry season months (October to April). This shows that the dry season is the best season for solar energy harvesting. During the rainy season (April to September), the amount of total radiation reduce notably compare with others. The annual averages of these parameters are shown in Table 6. Values obtained for global, beam and diffuse radiation, as well as clearness index, showed that all the sites considered in this study enjoy considerable solar energy potential suitable for solar energy conversion applications.

Furthermore, adjustment of solar panels on the optimum tilt angle and orientation as reflected in Table 7 plays a significant role in increasing the performance of PV panels. Monthly optimum tilt angle and the resulting solar energy gain (SEG) for the six selected locations are shown in the table. Monthly optimum tilt angle which corresponds to the maximum monthly solar radiation for Iseyin, Sokoto, Maiduguri, Jos, Enugu and Port-Harcourt lies

respectively in the range 0–39.8°, 0–44.5°, 0–44.1°, 0–43.2°, 0–38.5° and 0–36.3°. For all the sites, the optimum angle is equal to 0° in the month of April, May, June, July and August. This implies that the solar panels can be set in the same position (horizontally) for all these five months and still yield optimal result. However, due to low altitude of the Sun in January and December, the panel surface is expected to be adjusted at higher tilt angle if higher solar gain is desired. The results reveal that optimum tilt angle increases with increase in the latitude angle. In terms of SEG, highest values are obtained for Sokoto (32.2%), Maiduguri (31.4%), Jos (30.6%), Iseyin (23.4), Enugu (21%), and Port-Harcourt (18.2%); all these were obtained in the month of December. It is clearly seen that the values of SEG in some months for all the sites are very low. Thus, it can be concluded that if the solar panel surfaces are maintained in horizontal position between the months of March to September (rainy season) for all the sites, there would be no effect on the total solar gain. Since, monthly tilt angle adjustment may usually be impractical due to its cost, seasonal and yearly fixed

Table 7

Monthly, seasonal and annual mean variation of tilt angle and corresponding solar energy gain (SEG).

Site	Iseyin		Sokoto		Maiduguri		Jos		Enugu		Port Harcourt	
	β_{opt} (°)	SEG (%)	β_{opt} (°)	SEG (%)	β_{opt} (°)	SEG (%)	β_{opt} (°)	SEG (%)	β_{opt} (°)	SEG (%)	β_{opt} (°)	SEG (%)
Jan	36.3	18.1	41.6	26.6	41.6	27.0	40.4	25.6	34.5	15.9	33.3	10.1
Feb	26.1	8.6	31.5	13.7	31.3	13.7	29.8	12.4	24.7	7.6	23.0	6.5
Mar	11.0	1.4	15.9	3.10	15.6	3.1	13.6	2.3	9.2	0.9	7.6	0.6
Apr	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
May	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
Jun	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
Jul	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
Aug	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
Sep	3.5	0.1	8.2	0.8	8.0	0.8	5.7	0.4	2.5	0.1	1.0	0.0
Oct	20.2	4.7	26.4	9.2	26.4	9.3	23.2	6.7	19.4	4.4	16.8	3.1
Nov	34.7	16.8	39.6	24.0	39.5	24.0	37.7	21.5	33.3	15.3	30.1	11.4
Dec	39.8	23.4	44.5	32.2	44.1	31.4	43.2	30.6	38.2	21.0	36.3	18.2
Dry	29.6	11.4	34.6	16.6	34.4	16.5	32.9	15.4	28.0	10.1	26.1	7.8
Rainy	3.4	4.80	4.90	10.0	4.90	10.1	4.1	7.1	3.1	4.5	2.5	3.1
Annual	14.3	5.7	17.3	9.1	17.2	9.1	16.1	8.29	13.5	5.4	12.3	4.15

Table 8

Comparison of various system configurations in the selected sites.

Site	System Configuration	Components size					Economics		
		PV (kW)	Wind (kW)	Diesel (kW)	Batt. (unit)	Conv. (kW)	Initial capital (\$)	Total NPC (\$)	COE (\$/kWh)
Iseyin	PV–diesel–battery	5.0	–	3	20	3	23,935	42,430	0.476
	PV–wind–diesel–battery	4.0	1	3	10	2	21,190	43,003	0.483
	Diesel–battery	–	–	3	10	1	4145	61,393	0.689
	Diesel alone	–	–	3	–	–	600	81,095	0.911
Sokoto	PV–wind–diesel–battery	3.0	1	3	10	3	18,235	29,888	0.336
	PV–diesel–battery	5.0	–	3	20	3	23,935	36,804	0.413
	Wind–diesel–battery	–	2	3	10	2	12,390	38,274	0.430
	Diesel alone	–	–	3	–	–	600	81,095	0.911
Maiduguri	PV–wind–diesel–battery	3.5	1	3	10	3	19,835	34,863	0.391
	PV–diesel–battery	4.5	–	3	10	3	21,190	35,313	0.396
	Diesel–battery	–	–	3	10	1	4145	61,393	0.689
	Diesel alone	–	–	3	–	–	600	81,095	0.911
Jos	PV–wind–diesel–battery	3.0	1	3	10	3	18,235	28,830	0.324
	Wind–diesel–battery	3.5	2	3	10	2	12,390	34,569	0.388
	PV–diesel–battery	4.5	–	3	20	3	22,335	38,201	0.429
	Diesel alone	–	–	3	–	–	600	81,095	0.911
Enugu	PV–wind–diesel–battery	4.0	1	3	10	2	21,190	39,079	0.449
	PV–diesel–battery	5.5	–	3	20	3	25,535	41,833	0.470
	Diesel–battery	–	–	3	10	1	4145	61,393	0.689
	Diesel alone	–	–	3	–	–	600	81,095	0.911
Port Harcourt	PV–diesel–battery	5.0	–	3	20	3	11,235	29,888	0.482
	PV–wind–diesel–battery	3.5	2	3	10	2	26,335	47,066	0.528
	diesel–battery	–	–	3	10	1	4145	61,393	0.689
	Diesel alone	–	–	3	–	–	600	81,095	0.911

adjustment strategies can thus be used as an alternative option. Table 7, also shows the seasonal (dry and rainy season) and yearly fixed optimum tilt angles for each site. The maximum optimal angle occurs in the dry season, while the minimum occurs in the rainy season. It was observed that (just as obtained in monthly tilt angles), the seasonal and yearly optimum tilt angles increase with increase in the latitude angle.

Optimum system configurations

The optimal configuration assessment for the selected rural sites, were carried out based on data collected from each locations and system load profile in Table 2. The feasibility of hybrid renewable energy system for powering rural health clinic is based on the availability of the energy resources in the selected sites and life-time cost of the configurations. HOMER performed an hourly time series simulation for every possible system configuration on a yearly basis to determine the operational characteristics such as:

(i) annual electricity production, (ii) annual load served, (iii) excess electricity, (iv) renewable fraction and so on. The renewable energy sources and diesel generator were evaluated to determine the feasibility of the system. HOMER searched for optimum system configuration and component sizes that meet the load requirement at the lowest NPC and then presents the results.

Table 8, present optimal results of the selected sites. Based on these results, the best system configuration for both Iseyin and Port Harcourt sites is hybrid PV–diesel–battery system, while PV–wind–diesel–battery is considered optimal for the rest of the locations (Sokoto, Maiduguri, Jos and Enugu). The optimal configuration obtained for Jos has the lowest NPC and COE of all the sites considered; this is due to high solar and wind resources availability in the site as compared to other locations. Diesel-alone system is seen as worst configuration with highest NPC and COE. Apart from its low capital cost, it is also as a result of the high costs of fuel and maintenance associated with the configuration. Furthermore, this has demonstrated that configuration with lowest capital cost,

Table 9

Comparison of emission in optimal system configuration and diesel-only system in the selected sites.

Pollutant emission (kg/year)	Optimal system configuration						Diesel-only system
	Iseyin	Sokoto	Maiduguri	Jos	Enugu	Port Harcourt	All sites
CO ₂	889.3	342.6	521.7	258.9	977.9	938	9211
CO	2.20	0.85	1.29	0.64	2.41	2.32	22.7
UHC	0.24	0.09	0.14	0.07	0.27	0.26	2.52
PM	1.17	0.06	0.10	0.05	1.18	0.17	1.71
SO ₂	1.79	0.69	1.05	0.52	1.96	1.88	18.5
NO _x	19.6	7.55	11.5	5.70	21.5	20.7	202.9

may not necessarily be the type with lowest cost of energy. A diesel-battery configuration cost 45% more than the best optimal configuration in the entire sites. It is also noted that the COE of diesel-alone system (without battery) is more than twice that of the optimal configuration in all the sites. The addition of renewable sources and batteries to the diesel system is considered a good investment in terms of fuel savings and emission reduction.

Table 9 shows the annual pollutant emission from the best optimal configurations and the existing conventional diesel-alone system in each of the selected sites. These pollutant emissions includes: carbon dioxide (CO₂), carbon monoxide (CO), unburned hydrocarbon (UHC), particulate matter (PM), sulfur dioxide (SO₂) and nitrogen oxide (NO_x). The conventional diesel based power generation configuration produces 9211 kg/year of CO₂ in all the sites, while the optimal hybrid renewable system configurations emitted 889.3, 342.6, 521.7, 258.9, 977.9 and 938 kg/year of CO₂ at Iseyin, Sokoto, Maiduguri, Jos, Enugu and Port-Harcourt respectively. In addition, a total of 90, 96, 94, 97, 89 and 90 percent of CO₂ emission could be saved if a hybrid renewable energy system configuration is adopted at Iseyin, Sokoto, Maiduguri, Jos, Enugu and Port-Harcourt respectively.

The percentage reduction in pollutants emission is proportional to the rate of renewable energy penetration in the system. Hence, inclusion of renewable energy resources such as PV and wind in the hybrid system will be an economically viable option for providing electricity to rural health centers. The system will further help in abating the prevalent global warming, which occurred as a result of CO₂ emission into the atmosphere.

Conclusion

The potentials of wind and solar energies of six selected locations in Nigeria (Sokoto, Maiduguri, Jos, Iseyin, Enugu and Port-Harcourt), and the optimal configurations of hybrid renewable system for rural health clinic (RHC) applications in grid-unconnected rural villages within the selected domains, were investigated. The major summary of the study is as follows:

- (1) The monthly mean as well as the annual mean power density for Iseyin and Port-Harcourt fall into Class 1. In Enugu and Maiduguri, the wind power classification exists under Class 2. For Sokoto, the annual mean power density stand at 321.53 W/m². The average annual power density in Jos exists as 458.89 W/m², making the site to fall into Class 7 of the wind power classification. The classification places Sokoto and Jos in a high potential site for large-scale grid connected wind turbine applications and the rest of the selected locations as low wind potential sites.
- (2) Monthly optimum tilt angle for Iseyin, Sokoto, Maiduguri, Jos, Enugu and Port-Harcourt lies respectively in the range 0–39.8°, 0–44.5°, 0–44.1°, 0–43.2°, 0–38.5° and 0–36.3°. The optimum angle is equal to 0° in the month of April, May, June, July and August for all the locations. Hence, solar panels can be set in the same position (horizontally) for all these five months and still yield optimal result.

- (3) The abundance of wind and solar resources in the country create an ideal environment for inclusion of renewable energy systems, such as PV and wind in the design and implementation of standalone power supply systems to improve rural health delivery. The reliability of hybrid systems is found to be enhanced when solar, wind and diesel production are used together; the size of battery storage is also reduced because there is less dependence on one method of power production.
- (4) The PV/wind/diesel/battery hybrid system configuration is considered optimum for applications concerning RHC at Sokoto, Maiduguri, Jos and Enugu, while hybrid systems involving PV/diesel/battery is considered ideal for RHC at Iseyin and Port-Harcourt locations. Hence, the hybrid system configurations which meet the desired load depend largely on the renewable energy potential quality. For instance, the optimal configuration obtained for Jos has the lowest NPC and COE of all the sites considered, due to the high solar and wind resources availability compared to other locations; more than half of the total energy production of the optimal system configuration in this site, is found to be provided by the solar and wind generator. The wind system hybrid configuration is not considered best option for RHC applications at Iseyin, due to the relatively low wind speed. However, the simulation results shows that it is still better than diesel-only configuration.
- (5) The diesel-only system provides the highest COE (\$0.911/kWh), and emits 9211 kg of CO₂ per year in all the site considered. This is huge and will have adverse effect on the environment as well as the health of patients in the rural health facilities.
- (6) The overall results indicated that not only does the hybrid system configurations perform better than diesel-only simulation with regards to the NPC for all the six sites, but also displayed better performance in the categories such as electrical, fuel consumption and CO₂ reduction.

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